SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1703**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

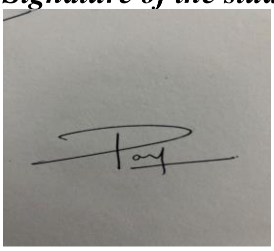
Annexure A

<u>Analytical Problem Solving</u>

Code of Conduct:

I PAVAN.P (192411017) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

<u>Analytical Problem Solving</u>

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

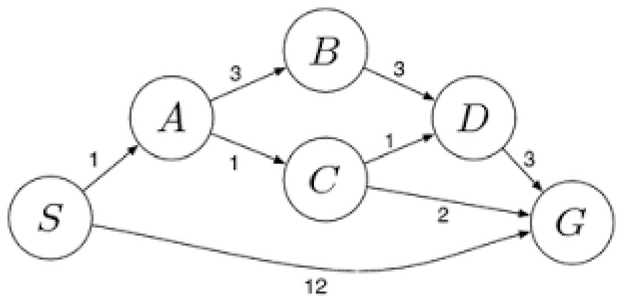
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

ANSWER FOR QUESTION NUMBER 1: -

Step	4-Gallon Jug	3-Gallon Jug	Action
Start	0	0	Initial state
1	0	3	Fill 3-gallon jug
2	3	0	Pour into 4-gallon jug
3	3	3	Fill 3-gallon jug again
4	4	2	Pour into 4-gallon until full
5	0	2	Empty 4-gallon jug
6	2	0	Pour remaining 2 gallons into 4-gallon jug

The **4-gallon jug contains exactly 2 gallons.**

ANSWER FOR QUESTION NUMBER 2: -

i. Percepts

- Camera images
- Rock and soil data
- Temperature
- Terrain information
- Obstacle detection
- Battery level

ii. Operating Environment

- Partially observable
- Sequential
- Dynamic
- Continuous
- Unknown environment

iii. Actions

- Move forward/backward
- Turn left/right
- Capture images
- Collect rock/soil samples
- Analyze samples
- Send data to Earth

iv. Performance Measure

- Successful exploration
- Accurate sample collection
- Safe navigation
- Energy efficiency
- Quality of transmitted data

v. Suitable Agent Architecture

Goal-Based + Utility-Based Agent

It plans routes, avoids obstacles, conserves power, and selects actions that maximize mission success

ANSWER FOR QUESTION NUMBER 3: -

Initial State

- Empty 8×8 chessboard

State Space

- All possible arrangements of 8 queens

Actions

- Place one queen in an empty column

Goal Test

- No two queens attack each other
    a. Same row
    b. Same column
    c. Same diagonal

Path Cost

- Each queen placement costs **1**
- Total solution cost = **8**

Search Strategy

- Backtracking Search
- Depth-First Search (DFS)

It plans routes, avoids obstacles, conserves power, and selects actions that maximize mission success

ANSWER FOR QUESTION NUMBER 4: -

Problem-Solving Agent

Goal-Based Agent

Reason: The goal is to reach the destination by selecting the most suitable cab.

Pseudocode

```
START
Input source and destination
Display available cab types
User selects cab
Check cab availability
IF cab available
   Calculate fare
   Confirm booking
   Assign driver
   Navigate to destination
ELSE
   Display "Cab not available"
END IF
STOP
```

ANSWER FOR QUESTION NUMBER 5:-

```
UCS(Start, Goal)

Create priority queue
Insert Start with cost 0

WHILE queue is not empty
   Remove node with lowest cost

   IF node is Goal
      Return path

   Expand node

   FOR each neighbor
```

NewCost = CurrentCost + EdgeCost
Insert neighbor with NewCost
END WHILE

Return Failure

FOR EXAMPLE: -

Step	Node	Cost
1	S	0
2	A	2
3	B	5
4	G (via A)	5

Least-Cost Path: S → A → G

Total Cost: 5

Always expands the node with the lowest cumulative path cost to find the optimal path.